

PROJECT DESIGN PHASE

Solution Architecture

Date	31 October 2025
Team ID	5C726FBC80DF9BA96244FB7712CD1C0D
Project Name	CRM Application for Jewel Management

1. Overview

The Solution Architecture for the CRM Application for Jewel Management is designed using Salesforce's multi-layered cloud infrastructure. It ensures seamless integration of data management, process automation, and analytics, enabling efficient jewellery business operations. The architecture follows a modular and layered design to support scalability, performance, and security.

2. Architectural Layers

A. Presentation Layer

- Built using Salesforce Lightning Experience, Lightning App Builder, and Lightning Pages.
- Provides intuitive navigation through custom tabs such as Jewel Customer, Item, Order, Price, and Billing.
- Ensures responsive, user-friendly access for Admin, Goldsmith, and Worker roles.
- Incorporates Dashboards and Reports for real-time visualization of sales, stock levels, and performance metrics.

B. Application Layer

- Contains the business logic implemented through Apex Classes and Trigger Handlers.
- Includes Record-Triggered Flows for automation of billing updates and stock adjustments.
- Validation Rules and Approval Processes ensure data consistency and integrity.

C. Data Layer

- Comprises custom and standard Salesforce objects such as Jewel Customer, Item, Price, Billing, and Customer Order.

- Relationships are built using Lookup and Master-Detail fields.
- SOQL Queries and Reports provide efficient data retrieval and analysis.

D. Integration Layer

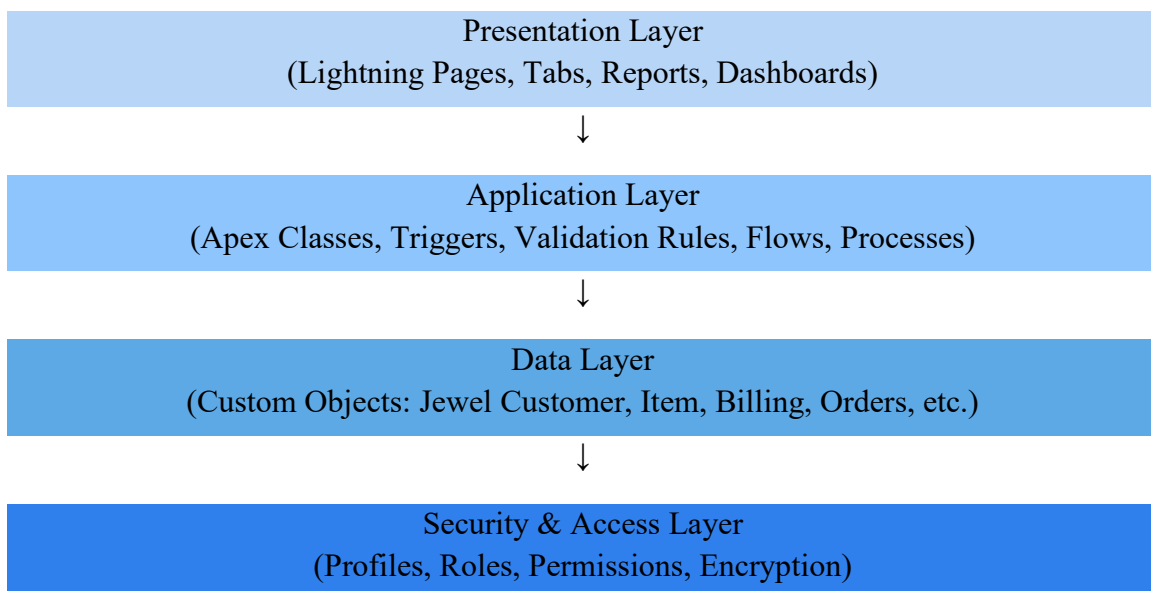
- Facilitates interaction between Reports, Dashboards, and Flows.
- Uses Process Builder and Flow Builder for cross-object automation.
- Supports future integration with ERP or e-commerce systems through Salesforce APIs.

E. Security Layer

- Implements Salesforce Role-Based Access Control (RBAC) and Profile Permissions.
- Roles include Admin, Goldsmith, and Worker, each with defined access levels.
- Includes field-level security, sharing rules, and encryption for secure operations.

3. System Flow Diagram

The following diagram represents the layered Salesforce architecture used in the system:

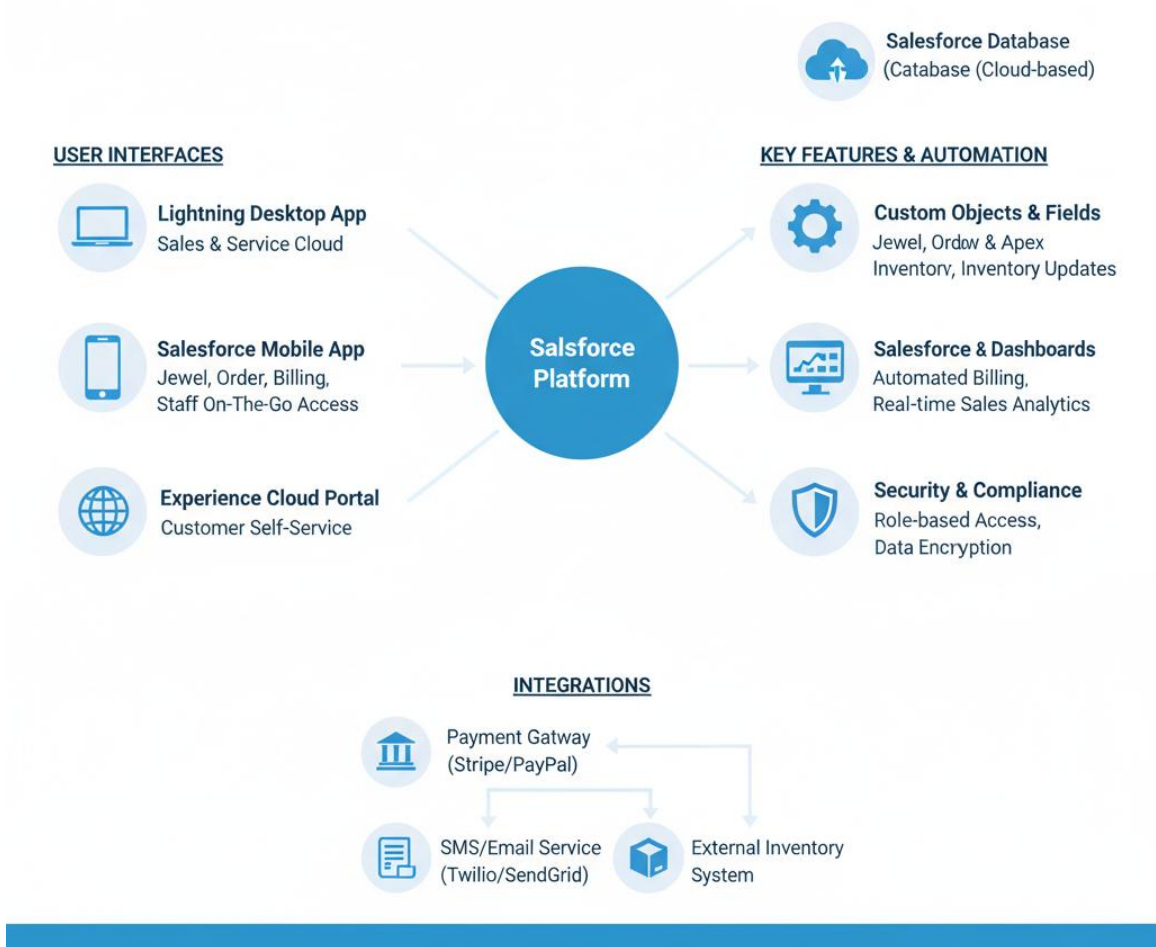


4. Architectural Highlights

- Cloud-Based: Entirely hosted on Salesforce Cloud, ensuring accessibility from anywhere.
- Scalable: New jewellery categories or users can be added without performance impact.
- Automated: All key workflows and approvals are automated.
- Data-Driven: Reports and Dashboards provide actionable insights.
- Secure: Controlled access and encryption protect business data.

JEWELLERY MANAGEMENT SYSTEM

Solution Architecture



5. Conclusion

The proposed Solution Architecture provides a reliable, scalable, and efficient framework for managing jewellery operations through Salesforce CRM. By integrating automation, data analytics, and role-based access, it ensures high performance, real-time visibility, and secure management of business processes.